

Project 2: Deploy Angular Application in Docker using Docker Compose (Development & Production)

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```
Z:\Om\SIT\Semester 7\DevOps Lab\Project 2>ng new angular-docker-demo

Would you like to share pseudonymous usage data about this project with the Angular Team
at Google under Google's Privacy Policy at https://policies.google.com/privacy. For more
details and how to change this setting, see https://angular.dev/cli/analytics.

No
Global setting: disabled
Local setting: No local workspace configuration file.
Effective status: disabled
✓ Which stylesheet system would you like to use? CSS [ https://developer.mozilla.org/docs/Web/CSS ]
✓ Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? Yes
✓ Which AI tools do you want to configure with Angular best practices? https://angular.dev/ai/develop-with-ai None
CREATE angular-docker-demo/.prettierrc (173 bytes)
CREATE angular-docker-demo/angular.json (2252 bytes)
CREATE angular-docker-demo/package.json (1048 bytes)
CREATE angular-docker-demo/README.md (1530 bytes)
CREATE angular-docker-demo/tsconfig.json (990 bytes)
CREATE angular-docker-demo/.editorconfig (331 bytes)
CREATE angular-docker-demo/.gitignore (666 bytes)
CREATE angular-docker-demo/tsconfig.app.json (464 bytes)
CREATE angular-docker-demo/tsconfig.spec.json (456 bytes)
CREATE angular-docker-demo/.vscode/extensions.json (134 bytes)
CREATE angular-docker-demo/.vscode/launch.json (490 bytes)
CREATE angular-docker-demo/.vscode/mcp.json (188 bytes)
CREATE angular-docker-demo/.vscode/tasks.json (1020 bytes)
```

```
CREATE angular-docker-demo/src/app/app.routes.server.ts (174 bytes)
CREATE angular-docker-demo/public/favicon.ico (15086 bytes)
✓ Packages installed successfully.
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
Successfully initialized git.
```

```
Z:\Om\SIT\Semester 7\DevOps Lab\Project 2>cd angular-docker-demo
```

```
Z:\Om\SIT\Semester 7\DevOps Lab\Project 2\angular-docker-demo>ng serve
```

Browser bundles

Initial chunk files	Names	Raw size
main.js	main	48.64 kB
styles.css	styles	95 bytes
	Initial total	48.74 kB

Server bundles

Initial chunk files	Names	Raw size
main.server.mjs	main.server	49.91 kB
server.mjs	server	1.84 kB
polyfills.server.mjs	polyfills.server	268 bytes

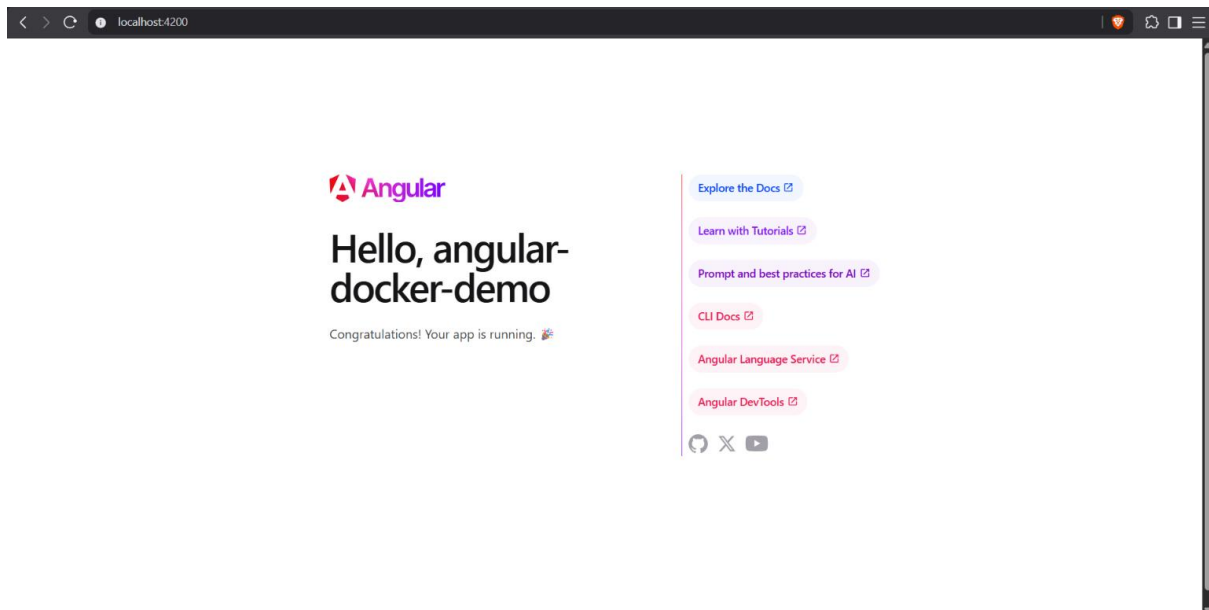
Application bundle generation complete. [6.284 seconds] - 2026-08-01T14:56:19.488Z

Watch mode enabled. Watching for file changes...

NOTE: Raw file sizes do not reflect development server per-request transformations.

→ Local: <http://localhost:4200/>

→ press h + enter to show help



```
1 FROM node:20
2
3 WORKDIR /app
4
5 COPY package*.json ./
6
7 RUN npm install
8
9 COPY . .
10
11 EXPOSE 4200
12
13 CMD ["npm", "start"]
```

```
2:\dev\Semester 7\DevOps Lab\Project 2\angular-docker-demo\docker-compose build
time="2026-08-01T20:37:26+05:30" level=warning msg="Z:\dev\Semester 7\DevOps Lab\Project 2\angular-docker-demo\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove
it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 631B done
#1 DONE 0.6s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 160B 0.0s done
#2 DONE 0.1s

#3 [internal] load metadata for docker.io/library/node:20
#3 DONE 2.8s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.6s

#5 [internal] load build context
#5 ...

#6 [1/5] FROM docker.io/library/node:20@sha256:8f693eaa7e0a8e71560c9a82b55fd54c2ae920a2ba5d2cde28bac7d1c01c9ba5
#6 resolve docker.io/library/node:20@sha256:8f693eaa7e0a8e71560c9a82b55fd54c2ae920a2ba5d2cde28bac7d1c01c9ba5 0.0s done
#6 DONE 0.2s

#5 [internal] load build context
#5 transferring context: 22.99MB 5.0s
#5 ...

#6 [1/5] FROM docker.io/library/node:20@sha256:8f693eaa7e0a8e71560c9a82b55fd54c2ae920a2ba5d2cde28bac7d1c01c9ba5
#6 sha256:0b22ec2d9b1a1a8e906e301b5ef08ef7774d5fc001b9cfa08e4e980152a0357 448B / 448B 0.3s done
#6 sha256:c3419f4430962505175b565eb8a24aafaa4d441c0b22234e94c6e8160848ba 3.32kB / 3.32kB 0.5s done
#6 sha256:76ebb204a89d20e7915fcd9a98ff33cb65fd7ce44ecf5a84b6cfa4bdb3d6d0 16.78MB / 48.62MB 6.8s
#6 sha256:2f2b064eb4d19f7b8506c38dc79c8a6f77acffb7622b015ceeb08fbc7380a1 1.25MB / 1.25MB 2.3s done
#6 sha256:2933ecab0ff11302fd71f29aa83ad2904681246f7a8320ad0dc3a60b23f05fee9 4.19MB / 211.55MB 6.5s
#6 sha256:8fda2b4d7993820b00c5488d173051e76d01ba6b85620617ba77001b0f9e2fa 1.05MB / 64.40MB 6.2s
#6 sha256:e5203b2bfeff92b72e816dcccbb1f1685f0cd592e521e3c0cfa195a78fe180e 7.34MB / 24.04MB 4.4s
#6 ...

#5 [internal] load build context
#5 ...

C:\WINDOWS\system32\cmd.exe - docker compose build
#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 160B 0.0s done
#2 DONE 0.1s

#3 [internal] load metadata for docker.io/library/node:20
#3 DONE 2.8s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.6s

#5 [internal] load build context
#5 ...

#6 [1/5] FROM docker.io/library/node:20@sha256:8f693eaa7e0a8e71560c9a82b55fd54c2ae920a2ba5d2cde28bac7d1c01c9ba5
#6 resolve docker.io/library/node:20@sha256:8f693eaa7e0a8e71560c9a82b55fd54c2ae920a2ba5d2cde28bac7d1c01c9ba5 0.0s done
#6 DONE 0.2s

#5 [internal] load build context
#5 transferring context: 22.99MB 5.0s
#5 ...

#6 [1/5] FROM docker.io/library/node:20@sha256:8f693eaa7e0a8e71560c9a82b55fd54c2ae920a2ba5d2cde28bac7d1c01c9ba5
#6 sha256:0b22ec2d9b1a1a8e906e301b5ef08ef7774d5fc001b9cfa08e4e980152a0357 448B / 448B 0.3s done
#6 sha256:c3419f4430962505175b565eb8a24aafaa4d441c0b22234e94c6e8160848ba 3.32kB / 3.32kB 0.5s done
#6 sha256:76ebb204a89d20e7915fcd9a98ff33cb65fd7ce44ecf5a84b6cfa4bdb3d6d0 16.78MB / 48.62MB 6.8s
#6 sha256:2f2b064eb4d19f7b8506c38dc79c8a6f77acffb7622b015ceeb08fbc7380a1 1.25MB / 1.25MB 2.3s done
#6 sha256:2933ecab0ff11302fd71f29aa83ad2904681246f7a8320ad0dc3a60b23f05fee9 4.19MB / 211.55MB 6.5s
#6 sha256:8fda2b4d7993820b00c5488d173051e76d01ba6b85620617ba77001b0f9e2fa 1.05MB / 64.40MB 6.2s
#6 sha256:e5203b2bfeff92b72e816dcccbb1f1685f0cd592e521e3c0cfa195a78fe180e 7.34MB / 24.04MB 4.4s
#6 ...

#5 [internal] load build context
#5 transferring context: 30.58MB 10.0s
#5 ...

#6 [1/5] FROM docker.io/library/node:20@sha256:8f693eaa7e0a8e71560c9a82b55fd54c2ae920a2ba5d2cde28bac7d1c01c9ba5
#6 sha256:76ebb204a89d20e7915fcd9a98ff33cb65fd7ce44ecf5a84b6cfa4bdb3d6d0 25.17MB / 48.62MB 13.7s
#6 sha256:2933ecab0ff11302fd71f29aa83ad2904681246f7a8320ad0dc3a60b23f05fee9 13.63MB / 211.55MB 13.5s
#6 sha256:8fda2b4d7993820b00c5488d173051e76d01ba6b85620617ba77001b0f9e2fa 6.29MB / 64.40MB 13.2s
#6 sha256:e5203b2bfeff92b72e816dcccbb1f1685f0cd592e521e3c0cfa195a78fe180e 17.83MB / 24.04MB 11.4s
#6 sha256:e5203b2bfeff92b72e816dcccbb1f1685f0cd592e521e3c0cfa195a78fe180e 19.92MB / 24.04MB 12.2s
#6 sha256:e5203b2bfeff92b72e816dcccbb1f1685f0cd592e521e3c0cfa195a78fe180e 22.02MB / 24.04MB 14.1s
#6 sha256:76ebb204a89d20e7915fcd9a98ff33cb65fd7ce44ecf5a84b6cfa4bdb3d6d0 28.31MB / 48.62MB 16.8s
#6 sha256:8fda2b4d7993820b00c5488d173051e76d01ba6b85620617ba77001b0f9e2fa 10.49MB / 64.40MB 17.1s
#6 sha256:2933ecab0ff11302fd71f29aa83ad2904681246f7a8320ad0dc3a60b23f05fee9 22.02MB / 211.55MB 10.6s
#6 sha256:e5203b2bfeff92b72e816dcccbb1f1685f0cd592e521e3c0cfa195a78fe180e 24.04MB / 24.04MB 17.1s done
#6 sha256:db53381ee51f9e43304e236099ba097ae1b33ae41a8007e0b6319992e55fd00 0B / 48.43MB 0.2s
```

```
C:\WINDOWS\system32\cmd.exe
#5 transferring context: 258.68MB 153.5s
#5 transferring context: 262.60MB 158.5s done
#5 DONE 158.6s

#8 [3/5] COPY package*.json ./
#8 DONE 0.3s

#9 [4/5] RUN npm install
#9 21.95 npm notice
#9 21.95 npm notice New major version of npm available! 10.8.2 -> 12.0.2
#9 21.95 npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.0.2
#9 21.95 npm notice to update run: npm install -g npm@12.0.2
#9 21.95 npm notice
#9 21.95
#9 21.95 added 485 packages, and audited 486 packages in 21s
#9 21.95
#9 21.95 115 packages are looking for funding
#9 21.95 run `npm fund` for details
#9 21.95
#9 21.95 3 moderate severity vulnerabilities
#9 21.95
#9 21.95 To address all issues (including breaking changes), run:
#9 21.95   npm audit fix --force
#9 21.95
#9 21.95 Run `npm audit` for details.
#9 DONE 22.3s

#10 [5/5] COPY . .
#10 DONE 6.0s

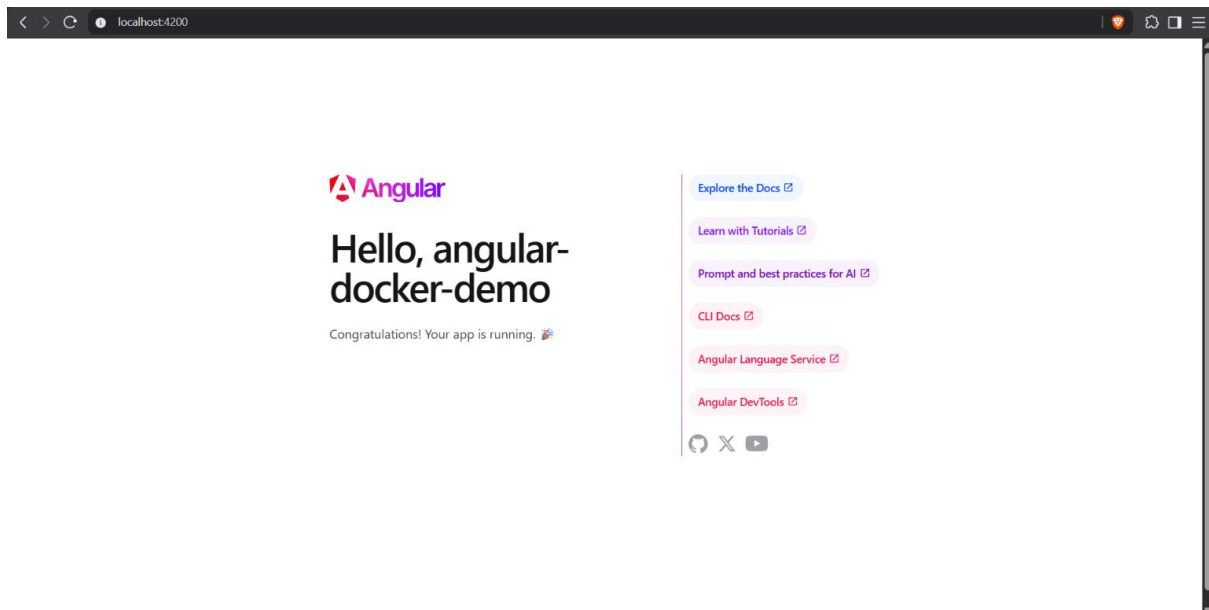
#11 exporting to image
#11 exporting layers
#11 exporting layers 15.0s done
#11 exporting manifest sha256:89899943601f86ce8167ce595e2356e81121f37bb5456533bcf55d16c0659e12 0.0s done
#11 exporting config sha256:a3a5fb71b3b91585257ef1dbdf6e115145184fc6d4241280ealab789d77946 0.0s done
#11 exporting attestation manifest sha256:a3f658e39936e1f822c346f29218ae8cd8021e4489d9a73f80eedc37add8ec 0.1s done
#11 exporting manifest list sha256:36807aeaf7176aab7d27108e4c64d18f03ae912d5a2dd4f2d66a85f9cc6eb0
#11 exporting manifest list sha256:36807aeaf7176aab7d27108e4c64d18f03ae912d5a2dd4f2d66a85f9cc6eb0 0.0s done
#11 naming to docker.io/library/angular-docker-demo-angular-app:latest done
#11 unpacking to docker.io/library/angular-docker-demo-angular-app:latest
#11 unpacking to docker.io/library/angular-docker-demo-angular-app:latest 10.0s done
#11 DONE 25.1s

#12 resolving provenance for metadata file
#12 DONE 0.1s
[+] build 1/1
✔ Image angular-docker-demo-angular-app Built 216.7s

Z:\Qm\Semester 7\DevOps Lab\Project 2\angular-docker-demo>
```

```
Z:\Qm\Semester 7\DevOps Lab\Project 2\angular-docker-demo>
Z:\Qm\Semester 7\DevOps Lab\Project 2\angular-docker-demo>docker compose up
time="2026-08-01T20:42:09+05:30" level=warning msg="Z:\Qm\Semester 7\DevOps Lab\Project 2\angular-docker-demo\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[*] up 2/2
✔ Network angular-docker-demo_default Created 0.1s
✔ Container angular-docker-demo-angular-app-1 Created 3.6s
Attaching to angular-app-1
angular-app-1 |
angular-app-1 | > angular-docker-demo@0.0.0 start
angular-app-1 | > ng serve
angular-app-1 |
angular-app-1 | B Building...
angular-app-1 | ✔ Building...
angular-app-1 | Browser bundles
angular-app-1 | Initial chunk files | Names | Raw size
angular-app-1 | main.js | main | 48.58 kB |
angular-app-1 | styles.css | styles | 95 bytes |
angular-app-1 | | Initial total | 48.67 kB |
angular-app-1 |
angular-app-1 | Server bundles
angular-app-1 | Initial chunk files | Names | Raw size
angular-app-1 | main.server.mjs | main.server | 49.84 kB |
angular-app-1 | server.mjs | server | 1.84 kB |
angular-app-1 | polyfills.server.mjs | polyfills.server | 268 bytes |
angular-app-1 | Application bundle generation complete. [3.637 seconds] - 2026-08-01T15:12:18.663Z
angular-app-1 |
angular-app-1 | Match mode enabled. Watching for file changes...
angular-app-1 | NOTE: Raw file sizes do not reflect development server per-request transformations.
angular-app-1 | 3:12:19 PM [vite] (css) Re-optimizing dependencies because vite config has changed
angular-app-1 | 3:12:19 PM [vite] (client) Re-optimizing dependencies because vite config has changed
angular-app-1 | → Local: http://localhost:4200/

View in Docker Desktop View Config Enable Match Detach
```



```
Dockerfile.prod > ...
1  # ----- Stage 1 : Build Angular App -----
2
3  FROM node:20 AS builder
4
5  WORKDIR /app
6
7  COPY package*.json ./
8
9  RUN npm install
10
11 COPY . .
12
13 RUN npm run build
14
15
16 # ----- Stage 2 : Nginx -----
17
18 FROM nginx:alpine
19
20 COPY nginx.conf /etc/nginx/conf.d/default.conf
21
22 COPY --from=builder /app/dist/browser /usr/share/nginx/html
23
24 EXPOSE 80
25
26 CMD ["nginx", "-g", "daemon off;"]
```

